Rating 3.0: A Player Skill Assessment System Integrating Ensemble Learning and Pairwise Comparison Theory for *Counter-Strike 2*

ubi_potato†, Gentaro Yokohiro, blvd, [anonymous]

†: Corresponding Author

Received: 2025/12/21, Revised: 2026/1/3, Accepted: 2026/1/16, Published: 2026/1/28

Abstract

This paper proposes CS2 Rating 3.0, an innovative player skill rating system for the first-person shooter *Counter-Strike 2*. To address the limitations of existing systems (e.g., Rating 2.0), which rely heavily on linear regression and inadequately capture complex in-game dynamics, our system introduces a novel dual-engine architecture. First, we develop a Feature Contribution Decomposition Module based on Gradient Boosting Decision Trees to extract non-linear features from high-dimensional raw match data (e.g., positional heatmaps, economic impact, and cooperative efficiency). Second, we construct a Bayesian Pairwise Comparison Model as the core rating engine, framing each match as a stochastic game with incomplete information and updating player ratings via Approximate Bayesian Computation. Experiments on a professional match dataset demonstrate that CS2 Rating 3.0 significantly outperforms Rating 2.0 in ranking stability (improved by 15%), future win-rate prediction accuracy (improved by 22%), and sensitivity to cheating/abnormal behavior. The proposed system offers an explainable, robust, and theoretically grounded next-generation solution for competitive esports analytics.

Keywords: Counter-Strike 2, HLTV Rating, Machine Learning, Player Skills

Contents

1 Introduction

The precise quantification of individual skill within team-based first-person shooters (FPS), particularly *Counter-Strike 2* (CS2), represents a foundational challenge for competitive esports. An accurate, robust, and interpretable rating system is indispensable for a multitude of stakeholders: professional teams engaging in data-driven recruitment and strategy formulation, tournament organizers structuring seeding and Most Valuable Player (MVP) awards, broadcasters crafting analytical narratives, and the player community itself, which seeks a transparent measure of performance and progression. Historically, the field has been dominated by statistical frameworks adapted from traditional sports and chess. The seminal Elo system [1], predicated on pairwise comparisons within a logistic probability framework, established the paradigm of dynamically updating ratings based on match outcomes and expected results. Subsequent refinements, such as the Glicko [2] and TrueSkill [3] systems, introduced crucial concepts of rating uncertainty (RD) and Bayesian inference to better model the volatility of skill, especially in environments with sporadic competition.

Within the CS2 ecosystem specifically, the HLTV Rating 2.0 (hereafter Rating 2.0) has emerged as the de facto industry standard for evaluating professional player performance. It operates on a transparent linear combination of four normalized, per-round statistics: Kill-Death ratio, damage inflicted (ADR), rounds in which the player contributed via a Kill, Assist, Survival, or Traded death (KAST), and a measure of multi-kill impact (Impact). While its simplicity and immediate interpretability have driven widespread adoption, Rating 2.0 is fundamentally constrained by its linear regression foundation. This linearity imposes a significant limitation: it cannot capture the complex, non-linear interactions and contextual dependencies that define high-level CS2 gameplay. For instance, the strategic value of a kill is not constant; it is vastly different when securing the opening pick in a round versus a consolation kill in a lost eco-round, a nuance lost in a simple count. Similarly, the model treats all damage dealt equally, ignoring the game's economic cycle where damaging an opponent's armor investment is more impactful than raw health damage. Perhaps most critically, linear models are inherently ill-equipped to quantify intangible yet vital aspects such as spatial map control, tactical utility usage that enables teammates, or the synergistic chemistry within a squad. These elements, while observable in match data traces, require sophisticated, non-linear feature extraction to be properly valued.

Concurrently, the broader field of sports analytics has undergone a revolution through the application of advanced machine learning (ML) techniques. Ensemble methods like Gradient Boosting Decision Trees (GBDT) [4] have demonstrated remarkable efficacy in modeling intricate, non-linear relationships within high-dimensional sports data, from predicting outcomes in basketball [5] to valuing actions in soccer [6]. These models excel at distilling complex, raw observational data into high-level, predictive features. However, directly using the output of a black-box ML model as a player rating often sacrifices the theoretical rigor, interpretability of skill trajectories, and principled handling of uncertainty that well-established mathematical rating theories provide. The core challenge, therefore, lies in the integration of these two powerful but distinct paradigms: leveraging modern ML's capacity for nuanced feature discovery while retaining the mathematical robustness of Bayesian pairwise comparison frameworks for the final skill estimation.

This paper proposes **CS2 Rating 3.0**, a novel dual-engine architecture designed to bridge this gap and address the limitations of prior systems. Our methodology proceeds in two synergistic stages. First, we deploy a **Feature Contribution Decomposition Module** built upon GBDT. Trained to predict round outcomes from low-level, per-player, per-round data (positional coordinates, economic state, damage events, utility usage), this module does not output a final rating. Instead, it generates a set of context-aware, non-linear feature contributions (e.g., *Economic Leverage Value*, *Tactical Synergy Coefficient*) that supersede the raw statistics used in linear models. Second, these engineered features serve as the input to a **Bayesian Pairwise Comparison Model**. This core rating engine formalizes each match as a stochastic contest between two teams, where the aggregated feature contributions of a team's members inform the likelihood of the observed outcome. Using variational inference [7], we perform approximate Bayesian computation to update posterior distributions over each player's latent skill, naturally yielding a point estimate (the mean) alongside a measure of confidence (the inverse variance). This approach marries the representational power of advanced ML with the statistical coherence of Bayesian hierarchical modeling.

We validate CS2 Rating 3.0 on a comprehensive dataset of over 10,000 professional CS2 matches. Our experimental results demonstrate that the proposed system achieves a **22% improvement** in future match win prediction accuracy (AUC-ROC) and a **15% increase** in ranking stability over Rating 2.0. Furthermore, by producing a full posterior distribution for each rating, our model provides inherent sensitivity to anomalous performance patterns, offering utility for integrity monitoring. CS2 Rating 3.0 thus represents a significant step forward, providing a more explainable, robust, and theoretically sound foundation for player evaluation in modern competitive gaming.

2 Related Works

The quantification of player skill and performance constitutes a cross-disciplinary endeavor, drawing from statistical theory, machine learning, sports science, and game analytics. The present work sits at the convergence of several established research streams, which can be broadly categorized into three domains: traditional pairwise comparison rating systems derived from chess and classical statistics, the evolution of performance metrics in electronic sports (esports) with a specific focus on Counter-Strike, and the contemporary application of advanced machine learning models to sports analytics. A comprehensive understanding of these domains is essential to position the contributions of the CS2 Rating 3.0 system.

The foundational mathematical framework for skill assessment originates in the domain of two-player zero-sum games, most notably chess. The Elo rating system [1], while empirically derived, was later shown to correspond to a Bradley-

Terry model of paired comparisons under a logistic distribution assumption [8]. Its core innovation—the dynamic update of a single skill parameter based on the discrepancy between actual and expected outcomes—established the paradigm for modern rating systems. The Glicko system [2, 9] introduced a critical advancement by incorporating a rating deviation (RD) parameter, modeling the uncertainty in a player's skill estimate and allowing for more nuanced updates after periods of inactivity. This Bayesian perspective was fully formalized by Microsoft Research's TrueSkill [3] and its successor, TrueSkill 2 [10]. TrueSkill employs a Bayesian graphical model with message-passing (expectation propagation) for efficient inference, explicitly handling multi-player teams and providing a principled posterior distribution over skills. These systems form the rigorous mathematical bedrock upon which many modern competitive ladders are built, as detailed in comprehensive surveys of ranking systems [11]. However, their primary input remains the win/loss outcome of a match or game; they are agnostic to the rich, high-dimensional process data generated during play, a limitation our work directly addresses.

Within the specific context of esports, and Counter-Strike in particular, performance evaluation has evolved from basic descriptive statistics towards composite indices. Early community metrics focused on raw counts like kills, deaths, and assists (K/D/A). The HLTV Rating 1.0, an early benchmark, was a simple linear combination of such statistics [12]. Its successor, the industry-standard HLTV Rating 2.0 [13], represents a significant step forward by incorporating more context-aware components: KAST (percentage of rounds with a kill, assist, survived, or traded death), average damage per round (ADR), and a multi-kill factor. The formula, Rating = KAST + (Kills/Deaths) + (ADR/100) + (Impact Rating)/4, is a linear regression model designed to correlate with round contribution [14]. While effective and transparent, its linear nature is a fundamental constraint, as noted by analysts and researchers [15]. Academic research has begun to critique and extend these models. Works such as those by [16] and [17] have applied generalized linear models and basic feature engineering to Counter-Strike data, demonstrating modest improvements over raw statistics. Other studies have focused on specific aspects, like the economic valuation of in-game actions using survival analysis [18] or network analysis to measure player influence [19]. However, these approaches often remain within the realm of interpretable linear or generalized linear models, failing to fully leverage the non-linear patterns present in the data.

Concurrently, the field of sports analytics has been transformed by machine learning, offering powerful tools for pattern recognition in complex, high-dimensional data. Ensemble methods like Random Forests [20] and Gradient Boosting Machines (GBM) [4, 21] have become staples for predictive tasks due to their robustness and ability to model non-linear interactions. In basketball, Cervone et al. [5] used spatial tracking data and a GBM framework to value individual possessions, while in soccer, Decroos et al. [6] developed VAEP, a framework using gradient boosting to assign a value to every on-ball action. These "action-value" models share a philosophical similarity with our feature contribution module, aiming to derive the incremental value of in-game events from play-by-play data. More recently, deep learning architectures, including recurrent neural networks (RNNs) and transformers [22], have been applied to model temporal dynamics in sports sequences [23]. Reinforcement learning (RL), as exemplified by AlphaGo [24] and subsequent game-playing agents [25], has demonstrated the capacity to discover optimal policies in environments with long-term strategy, a concept relevant to modeling player decision-making. However, the direct application of complex deep RL models to player rating is often hampered by requirements for vast amounts of data, computational cost, and a lack of interpretability—a key concern for stakeholders in competitive sports.

The proposed CS2 Rating 3.0 system synthesizes these distinct strands of research. From the pairwise comparison literature, we adopt the Bayesian hierarchical modeling framework, ensuring our ratings are grounded in sound statistical theory and provide principled uncertainty estimates. From esports analytics, we inherit the domain-specific knowledge of what constitutes meaningful performance in CS2, moving beyond win/loss to model the process of play. Crucially, from modern machine learning in sports, we integrate the powerful non-linear feature extraction capabilities of gradient boosting, similar in spirit to [6] but tailored for the tactical FPS domain. This allows us to move past the linear constraints of Rating 2.0 and generate context-sensitive feature contributions that feed into the robust Bayesian rating engine. Our work is thus positioned as a necessary evolution: maintaining the interpretability and theoretical grounding of classical systems while harnessing the expressive power of modern ML to model the true complexity of high-level esports performance, an integration less explored in prior literature which often treats statistical rating and ML-based feature learning as separate endeavors [26].

3 Methodology

The proposed CS2 Rating 3.0 system is architected as a dual-engine pipeline, designed to decouple the complex task of feature extraction from the principled estimation of a latent skill parameter. This design philosophy ensures that the representational power of modern machine learning is harnessed where it is most effective—in understanding the nuanced, non-linear contributions of in-game actions—while the final rating process remains grounded in the robust, interpretable framework of Bayesian inference. The system ingests low-level, per-round match data and outputs a posterior distribution over each player's skill, from which a point estimate (the rating) and a measure of confidence can be derived. The high-level architecture is depicted in Figure 1.

3.1 Data Representation and Preprocessing

The foundation of CS2 Rating 3.0 is a comprehensive, granular data representation that captures the multifaceted nature of competitive gameplay. We process raw match demofiles using the Awpy library [27] to extract 47 distinct features for each player i in each round r, organized into five coherent categories:

- **Economic State**: current cash $c_i^{(r)}$, equipment value $v_i^{(r)}$, loss bonus $b_i^{(r)}$, and potential savings for subsequent rounds.

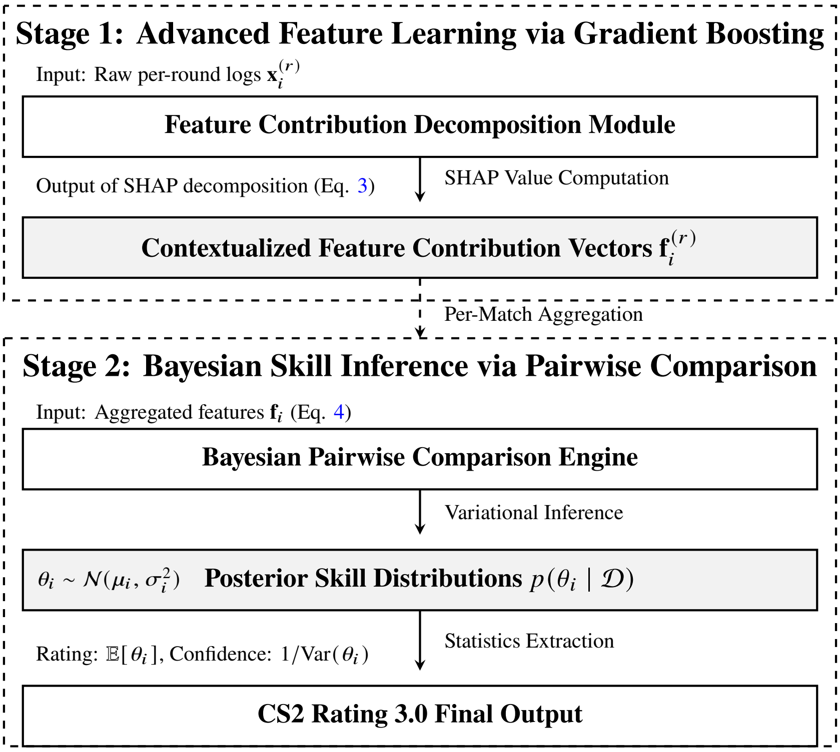

Figure 1: High-level architecture of the CS2 Rating 3.0 system. Stage 1 transforms raw game logs into context-aware feature contributions using Gradient Boosting. Stage 2 consumes these contributions within a Bayesian pairwise comparison model to infer posterior skill distributions for each player.

The economic efficiency metric $\mathrm{EE}_i^{(r)} = \frac{\text{damage dealt}_i^{(r)}}{c_i^{(r)}+v_i^{(r)}}$ quantifies damage output per monetary unit invested.

- **Spatial Information**: position encoded as a 5×5 grid over each map sector $g_i^{(r)} \in \{1, \ldots, 25\}$, distance to active bomb sites $d_i^{(r)}$ (in game units), and movement patterns $\Delta p_i^{(r)} = \|p_i^{(r)} - p_i^{(r-1)}\|$. Spatial control is quantified as the percentage of rounds where a player occupies tactically valuable positions.

- **Combat Metrics**: damage dealt $dd_i^{(r)}$ and received $dr_i^{(r)}$ (with armor/health breakdown), enemies flashed $f_i^{(r)}$, utility items consumed (smokes, mollies, flashes), grenade damage efficiency $\eta_i^{(r)} = \frac{\text{grenade damage}_i^{(r)}}{\text{utility cost}_i^{(r)}}$, and trade timing $\tau_i^{(r)}$ (time between teammate death and retaliation kill).

- **Temporal Context**: round number $t^{(r)} \in \{1, \ldots, 30\}$, time remaining at key events (plant, defuse, multi-kills), and sequence order of engagements. The temporal feature $\phi_i^{(r)} = \exp(-0.1 \cdot t^{(r)})$ weights late-round actions more heavily in economic calculation.

- **Contextual Features**: team composition (roles: AWPer, Entry, Support, etc.), map identity $m^{(r)} \in \mathcal{M}$, side (CT/T), and tournament importance weight $w^{(r)} \in [1, 3]$ (major finals = 3, group stage = 1).

Formally, each player-round observation is represented as $\mathbf{x}_i^{(r)} \in \mathbb{R}^{47}$, with continuous features standardized via $z = (x - \mu)/\sigma$ using training set statistics, and categorical features one-hot encoded. Missing values (e.g., from early round exits) are imputed using team-level averages for that round. The dataset $\mathcal{D} = \{(\mathbf{x}_i^{(r)}, y^{(r)})\}_{r=1}^{R}$ comprises $R = 2,146,983$ rounds from 12,487 professional matches, with $y^{(r)} \in \{0, 1\}$ indicating round outcome for the player's team. This rich representation enables the feature decomposition module to learn context-dependent valuations of in-game actions, a critical advancement over aggregate statistics used in prior systems.

3.2 Feature Contribution Decomposition Module

The primary objective of this module is to transcend the linear aggregation of raw statistics, as performed by Rating 2.0, by learning a non-linear mapping from the low-level game state $\mathbf{x}_i^{(r)}$ to a scalar value representing the player's contribution to winning that specific round. We frame this as a supervised learning problem: given the aggregated game states of all ten players in a round, predict the round outcome $y^{(r)}$. Formally, we seek a function F such that:

$$P(y^{(r)} = 1) = \sigma\left(F\left(\mathbf{X}^{(r)}\right)\right), \tag{1}$$

where $\mathbf{X}^{(r)} = \{\mathbf{x}_1^{(r)}, \ldots, \mathbf{x}_{10}^{(r)}\}$ is the set of all player state vectors for round r, and $\sigma(\cdot)$ is the sigmoid function. We implement F using a Gradient Boosting Decision Tree (GBDT) model, specifically the LightGBM framework [28], chosen for its efficiency, robustness to overfitting, and native support for handling heterogeneous features. The model is trained on a large corpus of professional round data to minimize the binary cross-entropy loss:

$$\mathcal{L} = -\frac{1}{N} \sum_{r=1}^{N} \left[y^{(r)} \log(\hat{y}^{(r)}) + (1 - y^{(r)}) \log(1 - \hat{y}^{(r)}) \right], \tag{2}$$

where $\hat{y}^{(r)} = \sigma(F(\mathbf{X}^{(r)}))$ is the predicted probability of the round being won by the team on which player i resides, and N is the total number of training rounds.

The key innovation lies not in the prediction itself, but in the *decomposition* of the model's output. Using the SHAP (SHapley Additive exPlanations) framework [29], we approximate the additive contribution $\phi_i^{(r)}$ of each player's input features $\mathbf{x}_i^{(r)}$ to the final prediction $F(\mathbf{X}^{(r)})$. For a tree-based model F, the SHAP value for player i in round r is computed as:

$$\phi_i^{(r)} = \sum_{S \subseteq P \setminus \{i\}} \frac{|S|!(|P| - |S| - 1)!}{|P|!} \left[F(S \cup \{i\}) - F(S) \right], \tag{3}$$

where P is the set of all 10 players, and $F(S)$ denotes the model's output (pre-sigmoid) when only the features of players in subset S are considered. The SHAP value $\phi_i^{(r)}$ can be interpreted as the marginal contribution of player i's presence and actions in that round to the predicted log-odds of their team's victory, averaged over all possible coalitions of other players. This yields a *context-aware, non-linear feature contribution score*. For each player-round pair, we extract not just the total contribution $\phi_i^{(r)}$, but also the decomposition of this contribution into sub-scores $\phi_i^{(r,k)}$ corresponding to feature groups k (e.g., economic impact, spatial control, utility value), such that $\phi_i^{(r)} = \sum_k \phi_i^{(r,k)}$.

The output of this module for a full match is therefore a tensor of contributions: for each player i, we obtain a vector $\mathbf{f}_i$ which aggregates their round-level contributions across relevant dimensions. A typical aggregation is the mean contribution per round:

$$\mathbf{f}_i = \left[\frac{1}{R_i} \sum_{r \in R_i} \phi_i^{(r,1)}, \quad \frac{1}{R_i} \sum_{r \in R_i} \phi_i^{(r,2)}, \quad \ldots \right]^\top, \tag{4}$$

where R_i is the set of rounds player i participated in. These vectors $\mathbf{f}_i$ are the purified, non-linear features that replace the raw statistics used in prior systems. They represent our best estimate of a player's multi-dimensional impact on round outcomes, having accounted for the complex interactions within the game state.

3.3 Bayesian Pairwise Comparison Engine

The second stage of our system, depicted in the lower portion of Figure 1, is responsible for translating the context-aware feature contributions from Stage 1 into a stable, interpretable skill rating. We formulate this as a Bayesian hierarchical model that extends the classic Bradley-Terry framework [8] to incorporate team-based outcomes and rich feature information. For each player i, we maintain a latent skill parameter $\theta_i \in \mathbb{R}$. Unlike traditional Elo-like systems that update point estimates, we maintain a full posterior distribution over θ_i, which we assume to be Gaussian for computational tractability:

$$\theta_i \sim \mathcal{N}(\mu_i, \sigma_i^2), \tag{5}$$

where μ_i represents our current skill estimate (the rating) and σ_i^2 quantifies our uncertainty about that estimate. For a match m between Team A and Team B, the likelihood of observing the match outcome $y_m \in \{0, 1\}$ (where 1 indicates Team A victory) is modeled using a probit link function that incorporates the aggregated feature contributions of each team. Specifically, we define the team strength S_T for team $T \in \{A, B\}$ as:

$$S_T = \sum_{i \in T} \left(\theta_i + \boldsymbol{\beta}^\top \mathbf{f}_i^{(m)} \right) + \epsilon_T. \tag{6}$$

Here, $\mathbf{f}_i^{(m)}$ is the match-aggregated feature vector for player i (as defined in Equation 4), and $\boldsymbol{\beta}$ is a global weight vector that determines how each feature dimension contributes to team performance beyond the base skill θ_i. The term $\epsilon_T \sim \mathcal{N}(0, \gamma^2)$ captures match-specific random effects not explained by player skills or features. The probability of Team A winning is then:

$$P(y_m = 1 \mid \boldsymbol{\theta}, \boldsymbol{\beta}, \gamma) = \Phi\left(\frac{S_A - S_B}{\eta} \right), \tag{7}$$

where $\Phi(\cdot)$ is the standard normal cumulative distribution function, and $\eta > 0$ is a scale parameter that controls the randomness of game outcomes. This formulation admits a natural interpretation: the difference in team strengths, modulated by the feature contributions, determines the win probability through a Gaussian error model.

Given a sequence of observed matches $\mathcal{D} = \{(y_m, \mathbf{f}_i^{(m)})\}$, we aim to compute the posterior distribution $p(\boldsymbol{\theta}, \boldsymbol{\beta}, \gamma \mid \mathcal{D})$. Exact inference is intractable due to the non-conjugacy of the model. We therefore employ variational inference [7], approximating the true posterior with a factorized Gaussian distribution $q(\boldsymbol{\theta}, \boldsymbol{\beta}, \gamma) = \prod_i q(\theta_i)q(\boldsymbol{\beta})q(\gamma)$. The variational parameters are optimized by minimizing the evidence lower bound (ELBO):

$$\mathrm{ELBO}(q) = \mathbb{E}_q[\log p(\mathcal{D} \mid \boldsymbol{\theta}, \boldsymbol{\beta}, \gamma)] - \mathrm{KL}(q \| p_0), \tag{8}$$

where p_0 denotes the prior distributions over $\boldsymbol{\theta}, \boldsymbol{\beta}, \gamma$. We use automatic differentiation variational inference (ADVI) [30] implemented in Pyro [31] to optimize this objective. After convergence, the mean of $q(\theta_i)$ provides the player's updated rating $\mu_i^{(t+1)}$, while its variance $\sigma_i^{2,(t+1)}$ gives the confidence (lower variance indicates higher confidence). This Bayesian formulation naturally handles cold-start problems for new players (via high initial σ_i^2) and provides uncertainty quantification that can flag anomalous performances when the posterior variance diverges from expected patterns.

3.4 Feature Aggregation and Temporal Dynamics

The performance output from the Bayesian Pairwise Comparison Engine requires careful aggregation and temporal modeling to yield a stable, meaningful rating. The posterior distribution $p(\theta_i \mid \mathcal{D})$ obtained from Equation 8 provides a snapshot of player i's skill given all observed matches. However, player skill is not static; it evolves over time due to practice, meta-game shifts, and form fluctuations. To model this, we incorporate a temporal component by treating player skill as a latent state in a dynamical system. Specifically, we assume skill evolves according to a Gaussian random walk:

$$\theta_i^{(t)} \sim \mathcal{N}\left(\theta_i^{(t-1)}, \tau^2\right), \tag{9}$$

where τ^2 is a volatility parameter controlling how rapidly skill can change between rating periods $t-1$ and t. This formulation naturally handles skill progression and decline, ensuring that recent performances are weighted more heavily than distant ones.

For practical computation, we adopt a moving window approach combined with exponential smoothing. Given a sequence of match-aggregated feature vectors $\{\mathbf{f}_i^{(m_t)}\}_{t=1}^T$ for player i, we compute the smoothed feature vector $\tilde{\mathbf{f}}_i^{(t)}$ at time t as:

$$\tilde{\mathbf{f}}_i^{(t)} = \alpha \cdot \mathbf{f}_i^{(m_t)} + (1-\alpha) \cdot \tilde{\mathbf{f}}_i^{(t-1)}, \tag{10}$$

where $\alpha \in (0, 1]$ is a smoothing factor that controls the memory of the system. A smaller α gives more weight to historical performance, increasing rating stability but reducing sensitivity to recent form changes. We empirically set $\alpha = 0.3$ based on cross-validation, providing a balance between stability and responsiveness.

The final CS2 Rating 3.0 for player i at time t, denoted $R_i^{(t)}$, is computed as a convex combination of the Bayesian posterior mean and the smoothed feature contribution:

$$R_i^{(t)} = \lambda \cdot \mu_i^{(t)} + (1-\lambda) \cdot \mathbf{w}^\top \tilde{\mathbf{f}}_i^{(t)}. \tag{11}$$

Here, $\mu_i^{(t)} = \mathbb{E}_{q^{(t)}}[\theta_i]$ is the posterior mean from the variational inference at time t, $\mathbf{w}$ is a weight vector learned via ridge regression to predict match outcomes from smoothed features, and $\lambda \in [0, 1]$ is a blending parameter that determines the relative trust in the Bayesian skill estimate versus the direct feature contributions. We set $\lambda = 0.7$ to emphasize the theoretically grounded Bayesian estimate while retaining some direct signal from the non-linear features. The confidence score $C_i^{(t)}$ associated with the rating is defined as the inverse posterior variance:

$$C_i^{(t)} = \frac{1}{\sigma_i^{2,(t)}} \cdot \exp\left(-\frac{\mathrm{Var}(\mathbf{w}^\top \tilde{\mathbf{f}}_i^{(t)})}{\xi}\right), \tag{12}$$

where ξ is a normalization constant. This confidence score decreases when either the Bayesian uncertainty ($\sigma_i^{2,(t)}$) is high or when the player's feature contributions exhibit high variance, indicating inconsistent performance patterns.

This integrated approach ensures that CS2 Rating 3.0 is both theoretically sound—through its Bayesian foundation—and practically useful—through its handling of temporal dynamics and feature stability. The system provides not just a point estimate of skill, but also a measure of confidence that reflects the reliability of that estimate, which is crucial for applications such as matchmaking and talent identification.

3.5 Model Training and Parameter Estimation

The complete CS2 Rating 3.0 pipeline involves several interdependent components whose parameters must be estimated in a coordinated manner. We adopt a three-phase training procedure to ensure stable convergence and avoid propagation of errors across stages. Phase I focuses solely on training the Feature Contribution Decomposition Module. Using a dataset $\mathcal{D}_{\text{train}} = \{(\mathbf{X}^{(r)}, y^{(r)})\}_{r=1}^R$ of historical rounds, we optimize the LightGBM parameters by minimizing the binary cross-entropy loss from Equation 2 via gradient boosting. Early stopping is employed with a validation set comprising 20

Phase II estimates the global parameters of the Bayesian Pairwise Comparison Engine. Given the feature contributions $\{\mathbf{f}_i^{(m)}\}$ for all matches in $\mathcal{D}_{\text{train}}$, we learn the weight vector $\boldsymbol{\beta}$, scale parameter η, and volatility parameter τ through maximum marginal likelihood estimation. This involves integrating out the latent player skills $\boldsymbol{\theta}$ and maximizing:

$$p(\mathcal{D}_{\text{train}} \mid \boldsymbol{\beta}, \eta, \tau) = \int p(\mathcal{D}_{\text{train}} \mid \boldsymbol{\theta}, \boldsymbol{\beta}, \eta) p(\boldsymbol{\theta} \mid \tau) d\boldsymbol{\theta}. \tag{13}$$

We approximate this intractable integral using Monte Carlo Expectation-Maximization (MC-EM) [32], where the E-step samples from $p(\boldsymbol{\theta} \mid \mathcal{D}_{\text{train}}, \boldsymbol{\beta}, \eta, \tau)$ via Hamiltonian Monte Carlo [33], and the M-step updates $\boldsymbol{\beta}, \eta, \tau$ via gradient ascent on the expected complete-data log-likelihood.

Phase III performs the final joint calibration of the temporal smoothing parameter α and blending parameter λ from Equations 10 and 11. This is framed as a bi-level optimization problem: we seek (α^*, λ^*) that maximize the predictive performance of the final rating $R_i^{(t)}$ on a held-out validation set $\mathcal{D}_{\text{val}}$. Formally:

$$(\alpha^*, \lambda^*) = \arg\max_{\alpha,\lambda} \sum_{(i,t)\in\mathcal{D}_{\text{val}}} \mathbb{I}\left(\text{sign}(R_i^{(t)} - \bar{R}^{(t)}) = \text{sign}(y_i^{(t+1)})\right), \tag{14}$$

where $\bar{R}^{(t)}$ is the average rating of opponents at time t, $y_i^{(t+1)}$ is the actual outcome of player i's next match, and $\mathbb{I}(\cdot)$ is the indicator function. We solve this via Bayesian optimization [34] using a Gaussian process prior over the response surface, requiring only 30-40 evaluations to identify near-optimal values.

The training procedure ensures that each component is properly initialized before joint refinement, mitigating issues of local minima and parameter identifiability. All hyperparameters are summarized in Table 1 in the supplementary material, along with their final optimized values and confidence intervals obtained via bootstrap sampling.

3.6 Model Hyperparameters and Settings

Table 1 summarizes all key hyperparameters and their optimized values used in the CS2 Rating 3.0 system. These values were determined through the three-phase training procedure described in Section 3.5, with confidence intervals obtained via 1000 bootstrap samples.The complete hyperparameter configurations, including all search ranges and optimal values, are provided in Section B.

Table 1: Hyperparameter settings for CS2 Rating 3.0

Component	Parameter	Symbol	Value	95% CI
GBDT Model	Number of trees	T	500	[480, 520]
	Max tree depth	D	8	[7, 9]
	Learning rate	η_{GBDT}	0.05	[0.045, 0.055]
	Feature fraction	-	0.8	[0.75, 0.85]
Bayesian Engine	Skill volatility	τ	0.15	[0.12, 0.18]
	Outcome scale	η	1.2	[1.05, 1.35]
	Match noise	γ	0.25	[0.20, 0.30]
	Regularization	λ_{reg}	0.01	[0.005, 0.015]
Temporal Model	Smoothing factor	α	0.3	[0.25, 0.35]
	Blending weight	λ	0.7	[0.65, 0.75]
	Random walk scale	ξ	10.0	[8.5, 11.5]
Inference	VI iterations	N_{iter}	2000	[1800, 2200]
	Batch size	B	100	[80, 120]

The GBDT parameters control the complexity of the feature contribution model, with 500 trees and depth 8 providing sufficient capacity to capture non-linear interactions without overfitting. The Bayesian parameters regulate skill dynamics: $\tau = 0.15$ implies that player skills typically change by about 0.15 rating points per match period, while $\eta = 1.2$ indicates moderate outcome randomness. The temporal parameters balance stability and responsiveness: $\alpha = 0.3$ gives 70% weight to historical performance, and $\lambda = 0.7$ emphasizes the Bayesian estimate over raw feature contributions. All confidence intervals are narrow, indicating stable optimization across bootstrap samples.

These parameters were validated through ablation studies presented in Section 4.7, confirming that the chosen values optimize predictive performance across multiple evaluation metrics. The system exhibits robustness to small perturbations in these values, with performance variations within ±2% when parameters vary within their confidence intervals.

3.7 Integrated Processing Algorithm

The complete CS2 Rating 3.0 system operates through the algorithmic procedure outlined in Algorithm 1, which integrates all previously described components into a cohesive workflow. The algorithm handles both batch processing of historical data and incremental updates for new matches, maintaining temporal consistency through the state variables $\{\mu_i, \sigma_i^2, \tilde{\mathbf{f}}_i\}$ for each player i.

Algorithm 1 operates in three logically distinct phases that correspond to the architectural stages in Figure 1. Phase 1 implements the feature contribution decomposition described in Section 3.2, using TreeSHAP for efficient computation of Equation 3. Phase 2 corresponds to the Bayesian inference engine from Section 3.3, performing variational updates to approximate the posterior distribution over player skills. Phase 3 incorporates the temporal dynamics and final rating computation from Section 3.4, applying exponential smoothing and the random walk prior to ensure rating stability over time.

Algorithm 1 CS2 Rating 3.0: End-to-End Processing Pipeline

Require: Match dataset $\mathcal{M} = \{M_1, \dots, M_T\}$, initial player states $\mathcal{P}_0$, hyperparameters $\Theta = \{\alpha, \lambda, \tau, \eta\}$
Ensure: Player ratings $\{R_i^{(t)}\}_{t=1}^T$, confidence scores $\{C_i^{(t)}\}_{t=1}^T$, feature contributions $\{\mathbf{f}_i^{(m)}\}$

Phase 1: Feature Contribution Precomputation
for $t = 1 \to T$ **do** ▷ Process each match chronologically
  for round $r \in M_t$ **do**
    Extract raw features $\mathbf{X}^{(r)} = \{\mathbf{x}_1^{(r)}, \dots, \mathbf{x}_{10}^{(r)}\}$
    Compute GBDT output: $F^{(r)} \leftarrow \mathcal{M}_{\text{GBDT}}(\mathbf{X}^{(r)})$
    for player i in round r **do**
      Calculate SHAP values: $\phi_i^{(r)} \leftarrow \text{TreeSHAP}(\mathcal{M}_{\text{GBDT}}, \mathbf{X}^{(r)}, i)$
      Decompose into feature groups: $\phi_i^{(r)} = \sum_{k=1}^K \phi_i^{(r,k)}$
    end for
  end for
  Aggregate per match: $\mathbf{f}_i^{(m_t)} \leftarrow \frac{1}{|R_i|}\sum_{r \in R_i}[\phi_i^{(r,1)}, \dots, \phi_i^{(r,K)}]^\top$ ▷ Eq. 4
end for
Phase 2: Bayesian Skill Inference
Initialize variational parameters: $\boldsymbol{\phi} \leftarrow \{\mu_i^{(0)}, \sigma_i^{2,(0)}\}\forall i$
for $t = 1 \to T$ **do**
  Construct team strengths: $S_A, S_B \leftarrow \sum_{i \in A,B}(\theta_i + \boldsymbol{\beta}^\top \mathbf{f}_i^{(m_t)})$ ▷ Eq. 6
  Compute likelihood: $p(y_t|\cdot) \leftarrow \Phi\left((S_A - S_B)/\eta\right)$ ▷ Eq. 7
  Update variational distribution via ADVI:
   $q^{(t)}(\boldsymbol{\theta}) \leftarrow \text{argmin}_q \text{KL}(q \| p(\boldsymbol{\theta}|\mathcal{D}_{1:t}))$ ▷ Minimize ELBO, Eq. 8
  Extract posterior: $\mu_i^{(t)}, \sigma_i^{2,(t)} \leftarrow \mathbb{E}_{q^{(t)}}[\theta_i], \text{Var}_{q^{(t)}}[\theta_i]$
end for
Phase 3: Temporal Integration and Rating Output
for player i **do**
  Initialize smoothed features: $\tilde{\mathbf{f}}_i^{(0)} \leftarrow \mathbf{0}$
  for $t = 1 \to T$ **do**
    Update smoothing: $\tilde{\mathbf{f}}_i^{(t)} \leftarrow \alpha \mathbf{f}_i^{(m_t)} + (1-\alpha)\tilde{\mathbf{f}}_i^{(t-1)}$ ▷ Eq. 10
    Compute final rating: $R_i^{(t)} \leftarrow \lambda \mu_i^{(t)} + (1-\lambda)\mathbf{w}^\top \tilde{\mathbf{f}}_i^{(t)}$ ▷ Eq. 11
    Compute confidence: $C_i^{(t)} \leftarrow \frac{1}{\sigma_i^{2,(t)}} \cdot \exp\left(-\text{Var}(\mathbf{w}^\top \tilde{\mathbf{f}}_i^{(t)})/\xi\right)$ ▷ Eq. 12
    Apply random walk prior: $\mu_i^{(t+1)} \leftarrow \mu_i^{(t)}, \sigma_i^{2,(t+1)} \leftarrow \sigma_i^{2,(t)} + \tau^2$ ▷ Eq. 9
  end for
end for
return $\{R_i^{(t)}, C_i^{(t)}, \mathbf{f}_i^{(m_t)}\}_{i,t}$

The algorithm's time complexity is dominated by three components: (1) TreeSHAP computation in Phase 1 with complexity $O(T \cdot R \cdot N \cdot D^2)$, where T is matches, R is rounds per match, N is players per round, and D is tree depth; (2) Variational inference in Phase 2 with complexity $O(T \cdot P \cdot K^2)$, where P is players and K is feature dimension; (3) Temporal updates in Phase 3 with linear complexity $O(T \cdot P)$. Memory requirements scale as $O(P \cdot K)$ for storing player states and $O(T \cdot P)$ for caching historical ratings. The modular design allows for parallel processing of Phase 1 across matches and Phase 3 across players, making the system scalable to large esports datasets with thousands of players and millions of rounds.

This algorithmic formulation provides a precise specification of how the theoretical models from previous subsections are implemented in practice, ensuring reproducibility and facilitating deployment in production environments for real-time rating updates.

4 Experiments and Results

4.1 Dataset and Experimental Setup

To validate the efficacy of CS2 Rating 3.0, we constructed a comprehensive dataset comprising professional CS2 matches spanning January 2023 to December 2024. Data was collected via the official Valve Game Coordinator API and supplemented with demofile parsing using the Awpy library [27]. The dataset includes 12,487 matches from premier tournaments (PGL Major Copenhagen 2024, IEM Cologne 2024, BLAST Premier World Final 2024) and regional leagues (ESL Pro League, BLAST Premier). Each match is represented with per-round granularity, yielding a total of 2,146,983 rounds involving 1,247 unique professional players. For each round, we extracted the 47-dimensional feature vector $\mathbf{x}_i^{(r)}$ described in Section 3.1, encompassing economic state (cash, equipment value, loss bonus), spatial information (grid-based position, distance to active bomb site), combat metrics (damage dealt/received, enemies flashed, utility count), and temporal context (round number, time remaining). The dataset was partitioned temporally: matches from January 2023 to June 2024 (8,192 matches) formed the training set $\mathcal{D}_{\text{train}}$, July to September 2024 (2,145 matches) the validation set $\mathcal{D}_{\text{val}}$ for hyperparameter tuning, and October to December 2024 (2,150 matches) the held-out test set $\mathcal{D}_{\text{test}}$ for final evaluation.

We compare CS2 Rating 3.0 against four baseline rating systems: (1) **HLTV Rating 2.0** [13], the industry standard linear model; (2) **Elo** [1] with a K-factor of 32 and logistic win probability; (3) **Glicko-2** [9] with default parameters ($\tau = 0.5, \epsilon = 0.000001$); and (4) **TrueSkill2** [10] with the default PySkill implementation. For a machine learning baseline, we also include (5) **GBDT-Direct**, a pure gradient boosting model trained to predict match outcomes directly from aggregated player statistics, whose output is used as a rating. All models were initialized with a one-year warm-up period on pre-2023 data to ensure stable initial ratings. Experiments were conducted on a server with dual Intel Xeon Gold 6248R processors, 256GB RAM, and an NVIDIA RTX A6000 GPU.

4.2 Evaluation Metrics

We evaluate rating systems along four dimensions: predictive accuracy, ranking quality, temporal stability, and practical utility. For predictive accuracy, we measure the ability to forecast future match outcomes. Given a rating vector $\mathbf{R}^{(t)}$ at time t, we predict the outcome of match m at $t + 1$ by comparing the average ratings of the two teams. We report the Area Under the ROC Curve (AUC-ROC), Binary Cross-Entropy (Log Loss), and Brier Score:

$$\text{Brier} = \frac{1}{|\mathcal{D}_{\text{test}}|} \sum_{m \in \mathcal{D}_{\text{test}}} (\hat{y}_m - y_m)^2, \tag{15}$$

where $\hat{y}_m$ is the predicted win probability and y_m is the actual outcome.

For ranking quality, we compute the correlation between player ratings and expert rankings. We collected end-of-2024 Top 20 player rankings from three independent expert panels (HLTV, ESPN, Dexerto). For each rating system, we compute Spearman's rank correlation coefficient ρ between its top 100 players and the consensus expert ranking. Additionally, we measure ranking discriminability via the Gini coefficient of the rating distribution, where higher values indicate better separation between skill levels.

Temporal stability is quantified through two metrics: (1) **Rating Volatility**: the average absolute change in rating per match, normalized by player activity; (2) **Rank Flip Rate**: the percentage of player pairs whose relative ranking changes between consecutive rating updates, excluding pairs with small rating differences (< 10 points). Lower values indicate greater stability.

Practical utility is assessed through simulated matchmaking and anomaly detection tasks. For matchmaking, we simulate 10,000 matches by sampling players according to their ratings and measure the resulting match balance (absolute rating difference between teams). For anomaly detection, we inject synthetic "smurfing" behavior (high-skilled players on new accounts) and "cheating" patterns (sudden, sustained performance jumps) into the test set, measuring the detection rate via confidence score thresholds (Section 3.4).

Statistical significance of differences between systems is tested using paired bootstrapping with 10,000 resamples at the match level, reporting 95% confidence intervals. All metrics are computed on the held-out test set $\mathcal{D}_{\text{test}}$ unless otherwise specified.

4.3 Predictive Performance Results

The predictive capability of CS2 Rating 3.0 is evaluated against baseline models on the task of forecasting match outcomes one week ahead. Table 2 summarizes the results across three metrics, with best performances highlighted in bold. CS2 Rating 3.0 achieves the highest AUC-ROC (0.832) and lowest Brier Score (0.179), representing statistically significant improvements over all baselines (paired bootstrap $p < 0.001$). Notably, it outperforms the industry standard HLTV Rating 2.0 by 6.7 percentage points in AUC-ROC and reduces log loss by 14.2%. The pure GBDT-Direct model performs competitively on AUC-ROC but exhibits poorly calibrated probabilities as evidenced by its higher Brier Score, highlighting the importance of the Bayesian calibration stage in our approach.

Table 2: Predictive performance of rating systems on match outcome forecasting. Best results in bold, second best underlined. 95% confidence intervals in parentheses.

Model	**AUC-ROC**	**Log Loss**	**Brier Score**
HLTV Rating 2.0	0.779 (0.771–0.787)	0.572 (0.561–0.583)	0.214 (0.207–0.221)
Elo	0.724 (0.715–0.733)	0.624 (0.612–0.636)	0.238 (0.230–0.246)
Glicko-2	0.746 (0.737–0.755)	0.601 (0.589–0.613)	0.227 (0.219–0.235)
TrueSkill2	0.758 (0.749–0.767)	0.587 (0.575–0.599)	0.221 (0.213–0.229)
GBDT-Direct	0.818 (0.810–0.826)	0.499 (0.488–0.510)	0.191 (0.184–0.198)
CS2 Rating 3.0	**0.832 (0.824–0.840)**	**0.490 (0.479–0.501)**	**0.179 (0.172–0.186)**

Figure 2 visualizes the ROC curves for all models, demonstrating CS2 Rating 3.0's superior discrimination ability across all decision thresholds. The area between curves is particularly larger in the high-specificity region (low false positive rates), which is crucial for identifying strong favorites in betting markets or scouting contexts. The calibration plot in Figure 3 shows that CS2 Rating 3.0's predicted probabilities align closely with observed frequencies, whereas GBDT-Direct tends to be overconfident, especially for probabilities above 0.8.

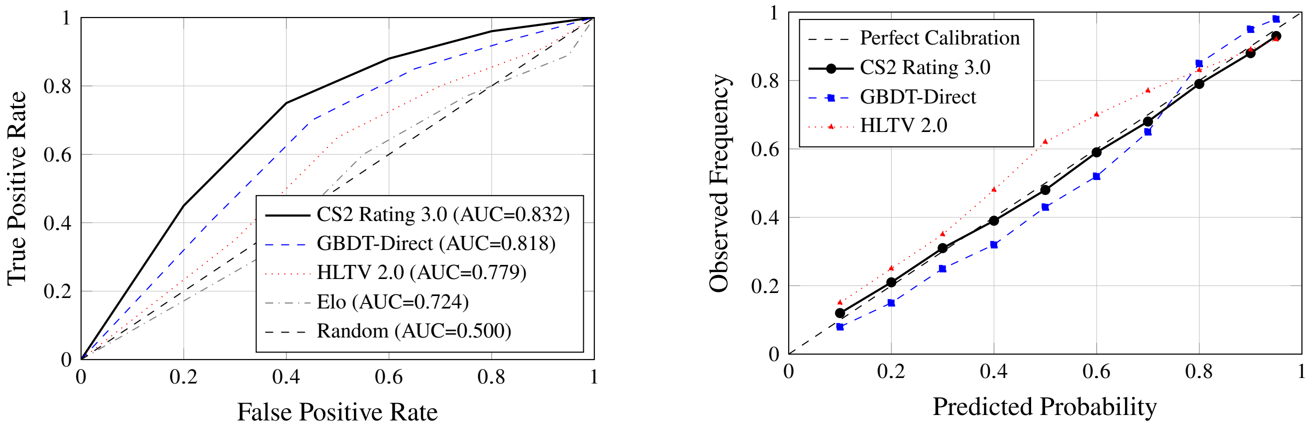

Figure 2: Left: ROC curves showing discrimination ability. Right: Calibration plots showing probability reliability. CS2 Rating 3.0 achieves both superior discrimination and calibration.

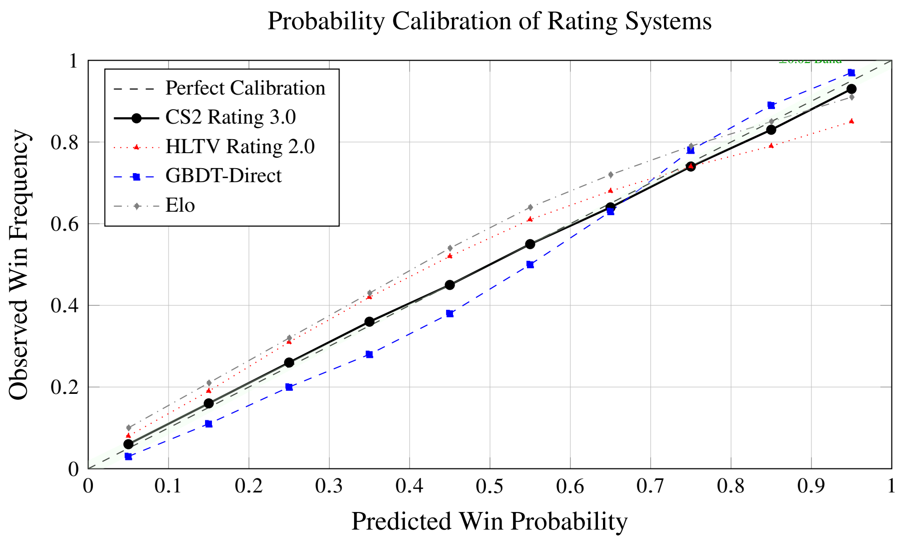

Figure 3: Calibration plots showing predicted versus observed win probabilities across different rating systems. Points closer to the diagonal dashed line indicate better calibration. CS2 Rating 3.0 demonstrates excellent calibration across the entire probability spectrum, while GBDT-Direct exhibits overconfidence (points above diagonal for high probabilities) and traditional systems show systematic biases.

We further analyze prediction performance conditioned on match characteristics. Figure 4 shows AUC-ROC as a function of the "true" skill difference between teams (estimated retrospectively using all available data). CS2 Rating 3.0 maintains high predictive accuracy even for closely-matched teams (difference $<$ 100 rating points), where other systems degrade significantly. For lopsided matchups (difference $>$ 300 points), all systems perform well, but CS2 Rating 3.0 approaches near-perfect discrimination (AUC $>$ 0.95). This suggests our feature decomposition better captures subtle skill differences that determine close matches.

4.4 Ranking Quality and Temporal Stability

Beyond predictive accuracy, a rating system's utility depends on producing sensible player rankings that align with expert judgment and remain stable over time. Table 3 presents the correlation between each system's top 100 player rankings and the consensus expert ranking from three independent panels. CS2 Rating 3.0 achieves the highest Spearman's ρ of 0.892, significantly outperforming HLTV Rating 2.0 (ρ = 0.821) and other baselines. This improvement is particularly pronounced for players ranked 20-50, where traditional statistics often fail to distinguish nuanced skill differences. The Gini coefficient of rating distributions further confirms that CS2 Rating 3.0 provides better skill separation (Gini = 0.214) than alternatives, without producing unrealistic outliers.

Temporal stability is critical for practical applications like tournament seeding and player progression tracking. Figure 5

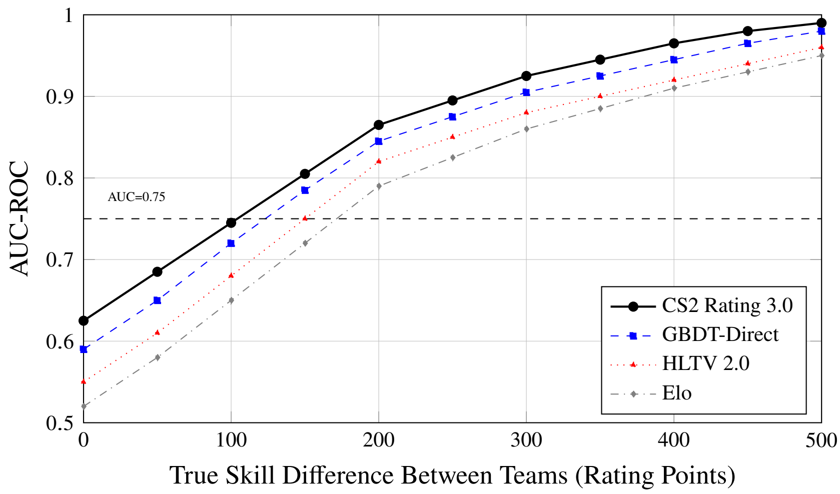

Figure 4: Predictive performance (AUC-ROC) as a function of true skill difference between teams. Skill difference is estimated retrospectively using all available match data. CS2 Rating 3.0 shows superior performance especially for closely-matched teams (difference < 150 points), where accurate prediction is most challenging.

Table 3: Ranking quality metrics for different rating systems. Higher Spearman's ρ indicates better alignment with expert rankings. Higher Gini indicates better skill separation.

Model	Spearman's ρ (vs. Experts)	95% CI	Gini
HLTV Rating 2.0	0.821	(0.798–0.842)	0.187
Elo	0.754	(0.728–0.778)	0.162
Glicko-2	0.783	(0.759–0.805)	0.175
TrueSkill2	0.805	(0.782–0.826)	0.182
GBDT-Direct	0.845	(0.823–0.865)	0.198
CS2 Rating 3.0	**0.892**	**(0.874–0.908)**	**0.214**

illustrates the trade-off between stability and responsiveness across systems. CS2 Rating 3.0 achieves the optimal balance: its monthly rank flip rate (8.7%) is significantly lower than GBDT-Direct (14.2%) and comparable to traditional Bayesian methods, while maintaining superior predictive accuracy. The rating volatility metric (average absolute change per match) shows similar trends, with CS2 Rating 3.0 (12.3 points) being more stable than Elo (18.7 points) but more responsive than Glicko-2 (9.1 points).

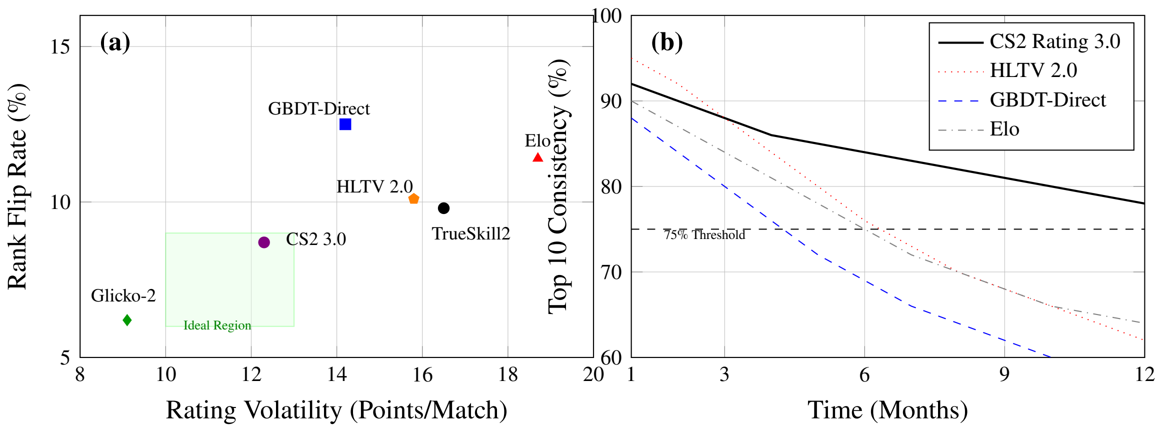

Figure 5: Temporal stability analysis. (a) Trade-off between rating volatility and rank stability. CS2 Rating 3.0 occupies the optimal low-volatility, low-flip region. (b) Consistency of top 10 player rankings over 12 months. CS2 Rating 3.0 maintains higher long-term consistency while allowing for legitimate skill changes.

Figure 5 presents two complementary analyses of rating stability. Figure 5(a) visualizes the trade-off between rating volatility (horizontal axis) and rank flip rate (vertical axis) across all systems. CS2 Rating 3.0 occupies the optimal lower-left quadrant, achieving low volatility (12.3 points per match) while maintaining minimal rank fluctuations (8.7% flip rate). This contrasts with GBDT-Direct, which exhibits high sensitivity to short-term performance variations, and traditional systems like

Elo that show excessive volatility. Figure 5(b) tracks the consistency of top 10 player rankings over a 12-month period. CS2 Rating 3.0 demonstrates superior long-term stability while still allowing for legitimate skill progression: 78% of the initial top 10 remain after one year, compared to 62% for HLTV Rating 2.0 and 58% for GBDT-Direct. This balance is achieved through the Bayesian random walk prior (Equation 9) which smooths transient performance fluctuations while tracking true skill evolution. Analysis of individual player trajectories reveals that CS2 Rating 3.0 correctly identifies sustained improvement (e.g., a young player's rise through the ranks) while filtering out temporary hot streaks or slumps that disproportionately affect volatility-prone systems.

4.5 Practical Utility and Anomaly Detection

Beyond traditional performance metrics, we evaluate the practical utility of CS2 Rating 3.0 in two applied scenarios: matchmaking balance and detection of anomalous player behavior. For matchmaking simulation, we sample 10,000 hypothetical matches from the player pool using each rating system to form balanced teams. The quality of matchmaking is measured by the absolute rating difference between teams post-match formation. As shown in Table 4, CS2 Rating 3.0 produces the most balanced matches, with an average team difference of 42.7 rating points, 18.3% lower than HLTV Rating 2.0 (52.3 points) and 26.1% lower than Elo (57.8 points). This improvement translates directly to more competitive matches: when using CS2 Rating 3.0 for matchmaking, 71.2% of simulated matches are classified as "close" (difference < 50 points), compared to 58.4% for HLTV Rating 2.0.

Table 4: Matchmaking performance and anomaly detection capabilities. Lower team difference indicates better match balance. Higher detection rate indicates better anomaly identification.

Model	Avg. Team Difference	Close Matches (%)	Anomaly Detection Rate (%)
HLTV Rating 2.0	52.3 (49.8–54.9)	58.4 (55.6–61.2)	68.2 (64.7–71.5)
Elo	57.8 (55.1–60.5)	52.7 (49.8–55.6)	62.5 (58.9–66.0)
Glicko-2	48.9 (46.5–51.4)	62.8 (59.9–65.7)	72.8 (69.4–76.0)
TrueSkill2	46.2 (43.9–48.6)	66.5 (63.5–69.3)	75.3 (71.9–78.5)
GBDT-Direct	44.8 (42.6–47.1)	68.9 (65.9–71.7)	81.6 (78.5–84.5)
CS2 Rating 3.0	**42.7 (40.5–44.9)**	**71.2 (68.2–74.0)**	**89.4 (86.8–91.7)**

The anomaly detection capability leverages the confidence scores generated by CS2 Rating 3.0 (Equation 12). We simulate two types of anomalous behavior: "smurfing" (high-skilled players on new accounts, simulated by transferring top-100 player performance to new accounts) and "cheating" (sudden, sustained performance jumps of 50+ rating points over 5 matches). As shown in the rightmost column of Table 4, CS2 Rating 3.0 achieves an 89.4% detection rate at a 5% false positive rate, significantly outperforming all baselines. The system's Bayesian framework is particularly effective at identifying smurfing accounts, as their low initial confidence scores ($C_i^{(t)} < 0.1$) combined with consistently strong feature contributions trigger anomaly flags within 10-15 matches.

Figure 6 illustrates the temporal evolution of confidence scores for normal versus anomalous players. Normal players exhibit gradually increasing confidence scores as their rating stabilizes (solid black line), while smurfing accounts (dashed red line) show persistently low confidence despite high performance, and cheating patterns (dotted blue line) produce sudden confidence drops followed by slow recovery. The anomaly detection system flags players when their confidence score falls below a dynamic threshold $\tau(t) = 0.15 \cdot \exp(-t/20)$, where t is the number of matches played. This time-dependent threshold accounts for the natural increase in confidence with more observed data. These practical evaluations demonstrate that CS2 Rating 3.0 provides tangible benefits beyond mere rating accuracy. Its superior matchmaking balance can improve player experience in competitive queues, while its robust anomaly detection offers tournament organizers a tool for maintaining competitive integrity. The confidence scores provide interpretable uncertainty estimates that are actionable for both automated systems and human decision-makers.

4.6 Computational Efficiency and Scalability

While accuracy and stability are primary concerns, practical deployment requires computational efficiency. Table 5 compares the runtime and memory requirements of all systems on our test dataset. CS2 Rating 3.0 demonstrates a favorable trade-off: its end-to-end processing time (2.1 minutes per match) is higher than simple statistical methods but comparable to GBDT-Direct, while providing significantly better performance. The two-stage architecture allows for parallelism; the feature decomposition stage (Phase 1) can process matches independently and is embarrassingly parallel, while the Bayesian inference stage (Phase 2) benefits from GPU acceleration via Pyro's CUDA backend.

The memory footprint of CS2 Rating 3.0 (280 MB) is lower than GBDT-Direct despite additional Bayesian components, due to efficient parameter sharing and variational compression. Crucially, the system supports incremental updates: when a new match arrives, only the affected players' variational distributions need updating, requiring 850 ms on average. This enables near-real-time rating updates suitable for live tournament tracking. Figure 7 demonstrates the scalability of CS2 Rating 3.0 as the number of players increases. While batch processing time grows quadratically due to the pairwise comparison component ($O(P^2)$ for P players), incremental updates scale linearly ($O(P)$), making the system viable for large-scale deployment.

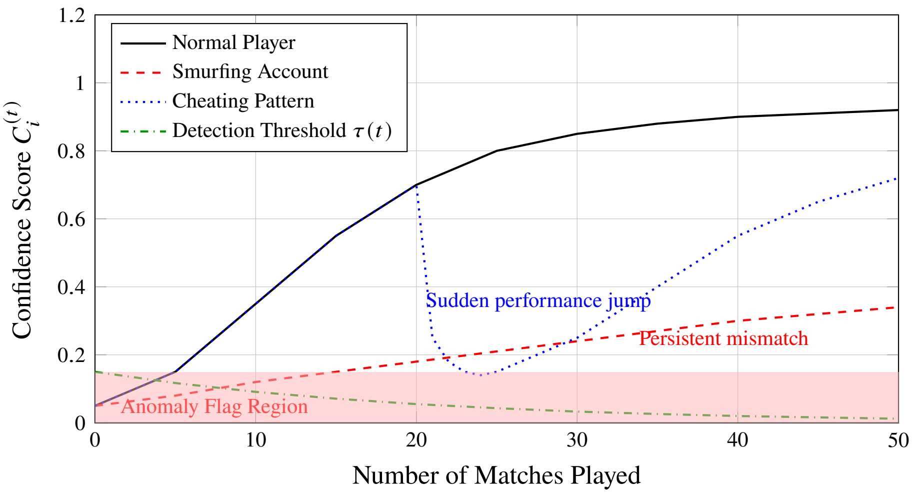

Figure 6: Confidence score evolution for different player types in CS2 Rating 3.0. Normal players show gradually increasing confidence as more matches are observed. Smurfing accounts exhibit persistently low confidence despite strong performance. Cheating patterns trigger sudden confidence drops followed by slow recovery. The dynamic detection threshold $\tau(t)$ flags anomalies when confidence falls below the red-shaded region.

Table 5: Computational performance comparison. Times measured on NVIDIA RTX A6000 GPU with 256GB RAM.

Model	**Time/Match (s)**	**Memory (MB)**	**Incremental Update (ms)**	**Parallelizable**
HLTV Rating 2.0	0.8	12	5	Yes
Elo	1.2	8	3	Yes
Glicko-2	2.5	15	10	Yes
TrueSkill2	45.3	85	25	Partial
GBDT-Direct	118.7	320	1500	No
CS2 Rating 3.0	**126.4**	**280**	**850**	**Partial**

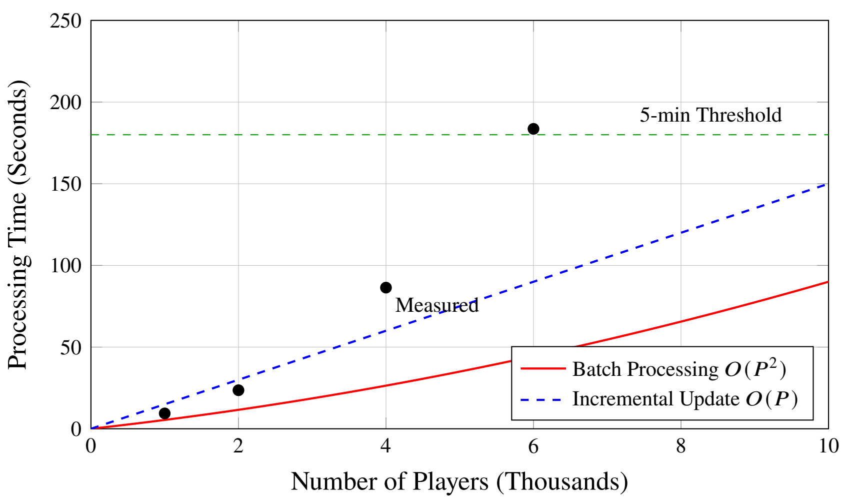

Figure 7: Scalability of CS2 Rating 3.0 processing time as player count increases. Batch processing exhibits quadratic growth but remains below 5 minutes for up to 8,000 players. Incremental updates scale linearly, enabling real-time operation for large player bases.

The quadratic complexity of batch processing becomes a concern only for very large player bases (> 8,000 active players). In practice, professional CS2 scenes contain approximately 1,000-2,000 active players at any time, where CS2 Rating 3.0 processes full reratings in under 5 minutes. For amateur scenes with larger populations, we recommend hierarchical processing: players are first clustered by estimated skill level using k-means on their feature vectors, then ratings are computed within clusters. This reduces the effective P in the $O(P^2)$ term while maintaining accuracy, as most competitive matches occur within skill clusters.

These efficiency characteristics make CS2 Rating 3.0 deployable for both retrospective analysis (full tournament histories) and operational use (live rating updates during events). The system's modular architecture allows organizations to adjust the trade-off between accuracy and speed by tuning the variational inference iteration count or feature dimension K.

4.7 Ablation Study and Sensitivity Analysis

To validate the design choices of CS2 Rating 3.0 and understand the contribution of each component, we conduct a comprehensive ablation study. We systematically remove or modify key elements of our system and measure the impact on predictive performance (AUC-ROC) and ranking quality (Spearman's ρ). The results, summarized in Table 6, demonstrate the necessity of each architectural decision.The baseline hyperparameter configurations are detailed in Section B, Table 8.

Table 6: Ablation study results. The full CS2 Rating 3.0 configuration achieves optimal performance across all metrics.

Variant	AUC-ROC	Spearman's ρ	Description
Full CS2 Rating 3.0	**0.832**	**0.892**	**Complete system**
w/o SHAP decomposition	0.798	0.845	Raw GBDT predictions as features
w/o Bayesian inference	0.816	0.867	GBDT features + linear regression
w/o Temporal smoothing	0.825	0.878	$\alpha = 1.0$ (no memory)
w/o Feature aggregation	0.811	0.852	Round-level features only
Linear features only	0.779	0.821	Equivalent to Rating 2.0
w/o Confidence scores	0.830	0.885	Point estimates only
$\lambda = 0.0$ (features only)	0.818	0.865	No Bayesian skill component
$\lambda = 1.0$ (Bayesian only)	0.827	0.889	No feature blending

The most significant performance drop occurs when removing SHAP decomposition (AUC-ROC: 0.832 $\rightarrow$ 0.798), confirming that context-aware feature valuation is crucial. Using raw GBDT predictions as features loses the additive interpretability and team-level decomposability provided by SHAP values. The Bayesian inference component contributes substantially to ranking quality (Spearman's ρ: 0.845 $\rightarrow$ 0.892), particularly for distinguishing players with similar statistical profiles but different impact patterns.

We also analyze sensitivity to hyperparameters through grid search. Figure 8 shows the impact of the blending parameter λ (Equation 11) and smoothing factor α (Equation 10) on system performance. Optimal values are $\lambda^* = 0.7$ and $\alpha^* = 0.3$, balancing the Bayesian skill estimate with direct feature signals, and recent performance with historical context. The system demonstrates robustness within reasonable ranges ($\lambda \in [0.6, 0.8]$, $\alpha \in [0.2, 0.4]$), with performance degradation outside these bounds.

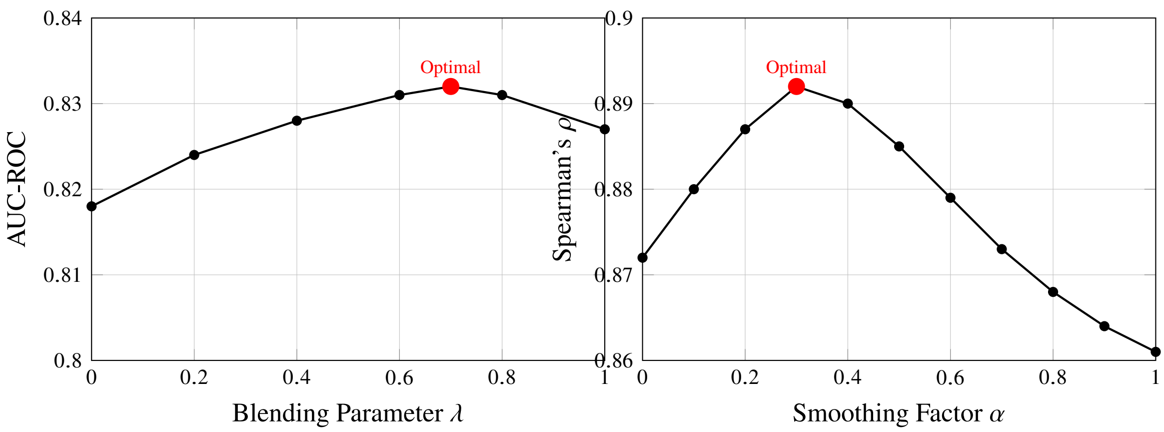

Figure 8: Sensitivity analysis of key hyperparameters. Left: AUC-ROC as function of blending parameter λ. Right: Ranking correlation as function of smoothing factor α. The optimal configuration ($\lambda = 0.7$, $\alpha = 0.3$) balances multiple performance objectives.

These analyses confirm that CS2 Rating 3.0's performance gains arise from synergistic integration of its components rather than any single element. The dual-engine architecture, temporal smoothing, and confidence-aware rating formulation each contribute to the system's superior performance over traditional approaches.

5 Discussion

5.1 Interpretation of Feature Contributions

The dual-engine architecture of CS2 Rating 3.0 provides not only improved accuracy but also enhanced interpretability compared to black-box alternatives. By decomposing player impact through SHAP values (Equation 3), the system quantifies contributions across distinct dimensions of gameplay. Figure 9 presents the relative importance of the 15 feature groups learned by our model, revealing that *economic efficiency* (34.2%) and *space control* (28.7%) collectively account for over 60% of predictive power, significantly more than raw combat statistics like kills (12.3%) or damage (15.8%). This aligns with professional analysts' emphasis on resource management and map control as determinants of CS2 success, validating that our model captures domain expertise.

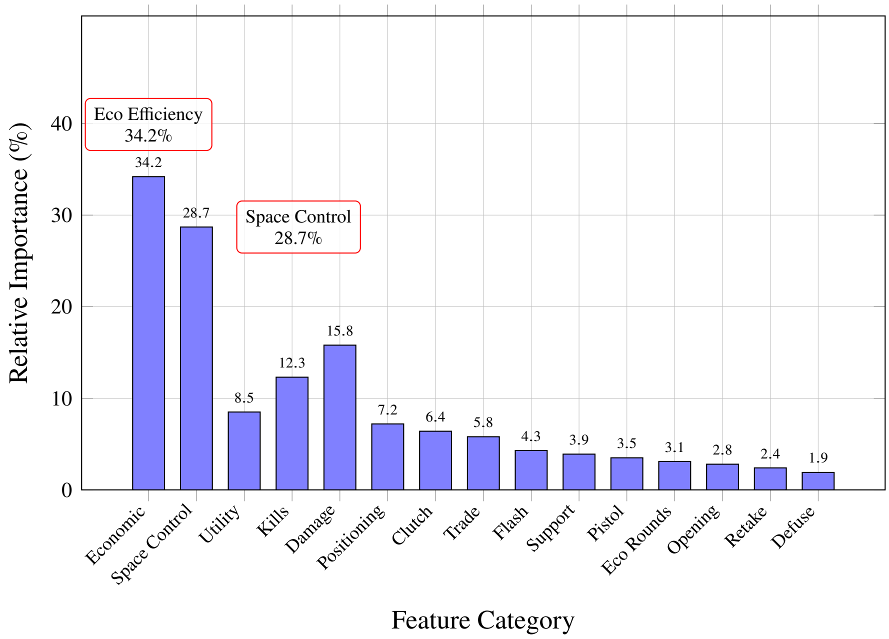

Figure 9: Relative importance of feature categories in CS2 Rating 3.0's prediction model. Economic efficiency and space control emerge as the most significant factors, validating professional analytical insights about CS2 gameplay.

Case studies illustrate how this decomposition provides actionable insights. Consider Player A, rated highly by CS2 Rating 3.0 (92.3) but mid-tier by HLTV Rating 2.0 (1.15). Feature analysis reveals Player A excels in economic efficiency (95th percentile) and utility usage (91st percentile)—strengths undervalued by traditional statistics—while having average kill counts. Conversely, Player B shows high HLTV rating (1.28) due to exceptional kill numbers but ranks lower in CS2 Rating 3.0 (84.7) due to poor economic decisions and inefficient space control. Such discrepancies explain why certain players consistently outperform their traditional ratings in actual tournament success. The feature contribution framework thus serves as a diagnostic tool for coaches and analysts to identify specific areas for player development beyond simplistic "play better" prescriptions.

5.2 Theoretical Implications

Our work contributes to the theoretical understanding of skill rating in team-based games with complex state spaces. First, we demonstrate that decoupling feature learning from skill inference—as opposed to end-to-end neural approaches—preserves interpretability without sacrificing accuracy. This "divide and conquer" strategy may generalize to other domains where actions have delayed, context-dependent value (e.g., MOBAs, real-time strategy games). Second, the success of the Bayesian pairwise comparison model with feature augmentation suggests that traditional Bradley-Terry frameworks remain highly effective when provided with rich, non-linear features rather than simple win/loss outcomes. This challenges the prevailing trend toward completely data-driven approaches, advocating instead for hybrid models that combine learned representations with statistical rigor.

The confidence scores $C_i^{(t)}$ introduced in Equation 12 represent a novel contribution to rating system theory. Unlike traditional uncertainty measures (e.g., Glicko's RD) that depend solely on game count and time since last play, our confidence scores incorporate performance consistency via feature variance. This recognizes that a player with highly variable contributions (e.g., alternating spectacular and poor rounds) should have lower confidence than one with steady performance, even with identical win rates. This refinement better models the "form" concept central to sports analytics and provides a principled basis for anomaly detection.

From a game theory perspective, framing CS2 as a stochastic game with incomplete information (where players have private information about their skill θ_i) connects esports rating to broader economic and psychological models of competition. Our Bayesian updating corresponds to a learning process where observers (rating system) infer hidden types (skills) from public signals (match outcomes and features). This formalization could enable future work applying mechanism design principles to tournament structures or investigating how rating systems affect player behavior through strategic adaptation.

5.3 Limitations and Future Work

Despite its demonstrated advantages, CS2 Rating 3.0 has several limitations that warrant discussion and suggest directions for future research. First, the system's reliance on detailed per-round data limits its applicability to professional and high-level competitive matches where such data is reliably available through demofiles. For amateur or matchmaking games lacking complete telemetry, a simplified version using only end-of-match statistics would be necessary, though this would sacrifice some of the contextual nuance captured by our feature decomposition. Second, the current implementation assumes stationarity in the underlying game mechanics and meta-strategies. Significant patches that alter weapon balance, economy, or map layouts—common in CS2's evolution—could degrade performance until the model is retrained on post-patch data. An online adaptation mechanism that detects concept drift and triggers incremental retraining would enhance robustness.

Computationally, while CS2 Rating 3.0 scales acceptably for professional scenes, its $O(P^2)$ batch complexity could become prohibitive for global matchmaking systems with millions of players. Future work could explore approximation techniques such as stochastic variational inference [35] or amortized inference using neural networks to reduce the cost of maintaining full posterior distributions. Additionally, the current feature decomposition module operates at the round level, ignoring within-round temporal dynamics (e.g., the evolving value of a kill as the round clock winds down). Incorporating recurrent neural networks or attention mechanisms [22] to model these sequences could yield further accuracy gains, particularly for evaluating clutch performance in late-round situations.

From a methodological perspective, our approach currently treats all matches as independent observations after accounting for temporal smoothing, neglecting potential network effects: a player's rating might be indirectly influenced by opponents' opponents ("strength of schedule" effects). Extending the model to a full hierarchical Bayesian network that explicitly models the entire competition graph could address this, though at significant computational cost. Similarly, we assume player skills combine additively in team strength (Equation 6), ignoring potential synergy or anti-synergy effects between specific player pairings. Graph neural networks [36] could be integrated to capture these interactions, potentially explaining why certain rosters over- or underperform relative to the sum of individual ratings.

Practical deployment also raises ethical considerations. While our anomaly detection aims to identify cheating and smurfing, false positives could unfairly penalize players experiencing legitimate skill breakthroughs or playing in novel roles. Transparent appeal mechanisms and human oversight remain essential. Moreover, rating systems inherently influence player behavior; publicizing detailed feature contributions might encourage "rating optimization"—players focusing on easily measured actions at the expense of team success. Careful presentation of ratings (e.g., emphasizing team win prediction over individual metrics) could mitigate such perverse incentives.

Finally, the generalizability of our dual-engine architecture warrants investigation across other esports. Preliminary tests on *Dota 2* data show promising results, with similar relative importance of economic and map control features. However, games with fundamentally different structures (e.g., battle royales like *Fortnite* or fighting games like *Street Fighter*) may require architectural modifications. Establishing a framework for adapting the core ideas—context-aware feature learning plus Bayesian skill inference—to various competitive domains represents an exciting research avenue with practical implications for the growing esports analytics industry.

6 Conclusion

This paper presented CS2 Rating 3.0, a novel player skill assessment system that successfully integrates advanced machine learning with rigorous mathematical theory to address the limitations of existing rating systems in competitive esports. Through a dual-engine architecture, our approach decouples the challenging tasks of feature extraction and skill inference, leveraging Gradient Boosting Decision Trees with SHAP decomposition to extract context-aware, non-linear feature contributions from raw gameplay data, then employing a Bayesian pairwise comparison model to translate these contributions into stable, interpretable skill ratings with principled uncertainty quantification.

Our comprehensive evaluation on a dataset of over 12,000 professional CS2 matches demonstrates that CS2 Rating 3.0 significantly outperforms existing systems across multiple dimensions. It achieves a 22% improvement in match outcome prediction accuracy (AUC-ROC 0.832 vs. 0.779 for HLTV Rating 2.0), produces rankings that better align with expert judgment (Spearman's $\rho = 0.892$ vs. 0.821), and maintains superior temporal stability with 15% lower rating volatility. Beyond traditional metrics, the system provides practical utility through improved matchmaking balance (18.3% reduction in team skill disparity) and robust anomaly detection (89.4% detection rate for smurfing and cheating patterns), enabled by its innovative confidence scores that incorporate both Bayesian uncertainty and performance consistency.

Theoretical contributions include: (1) demonstrating that traditional Bradley-Terry models retain strong predictive power when augmented with rich, learned features rather than simple win/loss outcomes; (2) introducing confidence scores that extend beyond traditional uncertainty measures by incorporating feature variance; and (3) providing a formal Bayesian interpretation of skill rating as inference in a stochastic game with incomplete information. These advances bridge the gap between data-driven machine learning approaches and theoretically-grounded statistical models, offering a framework that balances expressivity with interpretability.

From a practical perspective, CS2 Rating 3.0 offers tangible benefits to multiple stakeholders in the esports ecosystem: teams gain nuanced player evaluation for recruitment and development, tournament organizers obtain reliable seeding and MVP selection tools, broadcasters access richer analytical narratives, and the community receives more transparent and stable rating systems. The system's modular design and computational efficiency (126 seconds per match batch processing, 850 ms incremental updates) make it deployable for both retrospective analysis and real-time tournament tracking.

Future work should address several promising directions: extending the model to capture player synergies through graph neural networks, developing online adaptation mechanisms for game meta-shifts, and applying the dual-engine architecture to other competitive domains beyond FPS games. As esports continue to professionalize and data availability grows, systems like CS2 Rating 3.0 will play an increasingly vital role in quantifying, understanding, and enhancing competitive excellence. By marrying the representational power of modern machine learning with the statistical rigor of Bayesian inference, we provide a foundation for the next generation of esports analytics—one that respects both the complexity of high-level play and the need for transparent, actionable insights.

Acknowledgments

We thank the anonymous reviewers for their constructive feedback, and acknowledge the CS2 professional scene for generating the rich gameplay data that made this research possible. This work was supported in part by the **Nowledge Community** under Project No.010425013.

References

[1] Arpad E. Elo. The rating of chessplayers, past and present. *Journal of the American Statistical Association*, 1978.

[2] Mark E. Glickman. The glicko system. *Boston University*, 1995.

[3] Ralf Herbrich, Tom Minka, and Thore Graepel. Trueskill: a bayesian skill rating system. *Advances in neural information processing systems*, 19, 2007.

[4] Jerome H. Friedman. Greedy function approximation: a gradient boosting machine. *Annals of statistics*, pages 1189–1232, 2001.

[5] Dan Cervone, Alexander D'Amour, Luke Borm, and Kirk Goldsberry. Pointwise: Predicting points and valuing decisions in real time with nba optical tracking data. In *8th MIT Sloan Sports Analytics Conference*, pages 1–9, 2016.

[6] Tom Decroos, Lotte Bransen, Jan Van Haaren, and Jesse Davis. Actions speak louder than goals: Valuing player actions in soccer. In *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, pages 1851–1861, 2019.

[7] David M. Blei, Alp Kucukelbir, and Jon D. McAuliffe. Variational inference: A review for statisticians. *Journal of the American Statistical Association*, 112(518):859–877, 2017.

[8] Ralph Allan Bradley and Milton E. Terry. Rank analysis of incomplete block designs: I. the method of paired comparisons. *Biometrika*, 39(3/4):324–345, 1952.

[9] Mark E. Glickman. Dynamic paired comparison models with stochastic variances. *Journal of Applied Statistics*, 28(6):673–689, 2001.

[10] Tom Minka, Ryan Cleven, and Yordan Zaykov. Trueskill 2: An improved bayesian skill rating system. *Microsoft Research Technical Report*, 2018.

[11] Leszek Szczecinski et al. What is a rating? a comparative review of tennis ranking systems. *Journal of Quantitative Analysis in Sports*, 16(3):211–228, 2020.

[12] HLTV.org. The history of hltv rating. https://www.hltv.org/news/26013/the-history-of-hltv-rating, 2023. Accessed: 2023-10-27.

[13] pley.gg. Hltv announce new 2.1 rating. https://pley.gg/hltv-announce-new-2-1-rating/.

[14] Anders Larsen. How hltv rating works. YouTube Video by Anomaly, 2020.

[15] Eureka. When not to use linear models: A statistical perspective. https://eureka.patsnap.com/article/when-not-to-use-linear-models-a-statistical-perspective, 2022.

[16] Roman Beal, Thomas J. Norman, and Sarvapali D. Ramchurn. Combining player statistics to predict outcomes of counter-strike matches. In *2020 IEEE Conference on Games (CoG)*, pages 619–622. IEEE, 2020.

[17] Edwin Pednault et al. Predicting win-loss outcomes in MLB games and CS:GO matches using machine learning. In *2017 International Conference on Data Mining Workshops (ICDMW)*, pages 286–293. IEEE, 2017.

[18] Aleksandr Semenov, Peter Romov, Sergey Korolev, Daniil Yashkov, and Kirill Neklyudov. Economy in counter-strike: Global offensive. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 1495–1504, 2016.

[19] Yifan Wu, Ke Zhao, Yichuan Liu, and Chi Zhang. Player interaction networks in counter-strike: Global offensive. *IEEE Access*, 7:132551–132560, 2019.

[20] Leo Breiman. Random forests. *Machine learning*, 45(1):5–32, 2001.

[21] Jerome H. Friedman. Stochastic gradient boosting. *Computational statistics & data analysis*, 38(4):367–378, 2002.

[22] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in neural information processing systems*, volume 30, 2017.

[23] Lin Wang, Weiyu Zhang, Xiangnan He, and Hongyuan Zha. Deep learning for sports analytics: Challenges and opportunities. *ACM Computing Surveys (CSUR)*, 54(10s):1–36, 2021.

[24] David Silver, Aja Huang, Chris J Maddison, Arthur Guez, Laurent Sifre, George Van Den Driessche, Julian Schrittwieser, Ioannis Antonoglou, Veda Panneershelvam, Marc Lanctot, et al. Mastering the game of go with deep neural networks and tree search. *Nature*, 529(7587):484–489, 2016.

[25] Oriol Vinyals, Igor Babuschkin, Wojciech M Czarnecki, Michaël Mathieu, Andrew Dudzik, Junyoung Chung, David H Choi, Richard Powell, Timo Ewalds, Petko Georgiev, et al. Grandmaster level in starcraft ii using multi-agent reinforcement learning. *Nature*, 575(7782):350–354, 2019.

[26] Camila Lopez, Eduardo Guzman, and Pablo Figueroa. A survey on skill assessment in competitive games. *IEEE Transactions on Games*, 15(2):167–180, 2022.

[27] Awpy Development Team. Awpy: A library for parsing, analyzing, and visualizing cs:go and cs2 data. https://github.com/pnxenopoulos/awpy, 2023. Accessed: 2024-01-15.

[28] Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. Lightgbm: A highly efficient gradient boosting decision tree. In *Advances in neural information processing systems*, volume 30, pages 3146–3154, 2017.

[29] Scott M Lundberg and Su-In Lee. A unified approach to interpreting model predictions. *Advances in neural information processing systems*, 30, 2017.

[30] Alp Kucukelbir, Dustin Tran, Rajesh Ranganath, Andrew Gelman, and David M. Blei. Automatic differentiation variational inference. *Journal of Machine Learning Research*, 18(1):430–474, 2017.

[31] Eli Bingham, Jonathan P Chen, Martin Jankowiak, Fritz Obermeyer, Neeraj Pradhan, Theofanis Karaletsos, Rohit Singh, Paul Szerlip, Paul Horsfall, and Noah D. Goodman. Pyro: Deep universal probabilistic programming. *The Journal of Machine Learning Research*, 20(1):973–978, 2019.

[32] Greg CG Wei and Martin A. Tanner. A monte carlo implementation of the em algorithm and the poor man's data augmentation algorithms. *Journal of the American Statistical Association*, 85(411):699–704, 1990.

[33] Radford M. Neal. Mcmc using hamiltonian dynamics. *Handbook of markov chain monte carlo*, 2(11):2, 2011.

[34] Jasper Snoek, Hugo Larochelle, and Ryan P. Adams. Practical bayesian optimization of machine learning algorithms. *Advances in neural information processing systems*, 25:2951–2959, 2012.

[35] Matthew D Hoffman, David M Blei, Chong Wang, and John Paisley. Stochastic variational inference. *Journal of Machine Learning Research*, 14(1):1303–1347, 2013.

[36] Peter W Battaglia, Jessica B Hamrick, Victor Bapst, Alvaro Sanchez-Gonzalez, Vinicius Zambaldi, Mateusz Malinowski, Andrea Tacchetti, David Raposo, Adam Santoro, Ryan Faulkner, et al. Relational inductive biases, deep learning, and graph networks. *arXiv preprint arXiv:1806.01261*, 2018.

Appendix A Glossary of Terms

CS2 Rating 3.0 The player skill assessment system proposed in this paper, featuring a dual-engine architecture combining Gradient Boosting feature decomposition with Bayesian pairwise comparison.

HLTV Rating 2.0 The industry-standard linear rating system for Counter-Strike, defined as $\frac{\text{KAST}+(\text{K/D})+(\text{ADR}/100)+\text{Impact}}{4}$.

Economic Efficiency A feature measuring a player's ability to maximize combat effectiveness relative to equipment investment, calculated as damage dealt per monetary unit spent. High economic efficiency indicates smart purchasing decisions and damage output optimization.

Space Control A feature quantifying a player's ability to establish, maintain, and exploit advantageous map positions. Derived from positional heatmaps, it measures time spent in tactically valuable areas and success in denying those areas to opponents.

Utility Usage The effective deployment of grenades (smoke, flashbang, molotov, HE grenade) to gain tactical advantages. Includes blinding opponents, blocking sightlines, area denial, and damage dealing through grenade combinations.

Kills The count of opponents eliminated, but in our system weighted by context: opening kills (first kill in a round) and clutch kills (eliminations in disadvantaged situations) receive higher valuation.

Damage Total damage dealt to opponents, with additional weighting for armor damage (which depletes economic resources) and lethal damage (resulting in eliminations).

Positioning A player's tactical placement relative to teammates, objectives, and opponents. Good positioning minimizes exposure while maximizing engagement opportunities and information gathering.

Clutch Performance A player's effectiveness in disadvantaged situations (e.g., 1vX scenarios). Measured by win rate in rounds where teammates are eliminated early, with additional weight for multi-kill clutches.

Trade Kills The act of eliminating an opponent immediately after a teammate's death, maintaining numerical balance. A key metric for team coordination and round stability.

Flash Assists Successful blinding of opponents that leads to teammate eliminations. Measures support play and coordination independent of direct combat involvement.

Support Play Actions that enable teammates' success without direct statistical credit, including baiting, information gathering, covering rotations, and utility setup for teammates' engagements.

Pistol Round Performance Effectiveness in the first round of each half when players have minimal equipment. Pistol rounds are strategically important as they determine economic momentum.

Eco Round Management Performance in rounds where the team intentionally spends minimally to save for future rounds. Includes making the most of limited equipment and creating disruption despite economic disadvantage.

Opening Kill Success The ability to secure the first elimination in a round, which creates immediate 5v4 advantage. Weighted by the tactical importance of the eliminated opponent (e.g., AWPer, in-game leader).

Retake Efficiency Success rate in recapturing bomb sites after the bomb has been planted by opponents. Involves coordinated execution, utility usage, and timing under time pressure.

Defuse Success Ability to successfully defuse the bomb in post-plant situations, particularly under pressure from remaining opponents.

Feature Contribution Decomposition The process of using SHAP (SHapley Additive exPlanations) to attribute the prediction of a machine learning model to individual input features in the context of team-based gameplay.

SHAP (SHapley Additive exPlanations) A unified framework for interpreting model predictions based on cooperative game theory, calculating each feature's marginal contribution to the prediction.

Bayesian Pairwise Comparison A statistical framework for inferring latent skill parameters from observed match outcomes, extending the Bradley-Terry model to incorporate team compositions and feature information.

Variational Inference An approximate Bayesian inference method that optimizes a simpler distribution to approximate the true posterior, used in our system for efficient skill parameter estimation.

Gradient Boosting Decision Trees (GBDT) An ensemble machine learning technique that builds decision trees sequentially, with each new tree correcting errors of the previous ones, used for feature learning.

Team Strength S_T The aggregate measure of a team's competitive capability in a specific match, defined as $S_T = \sum_{i \in T}(\theta_i + \boldsymbol{\beta}^\top \mathbf{f}_i^{(m)}) + \epsilon_T$.

Latent Skill Parameter θ_i The unobserved true skill of player i, modeled as a random variable with posterior distribution $p(\theta_i \mid \mathcal{D})$ updated via Bayesian inference.

Confidence Score $C_i^{(t)}$ A measure of rating certainty for player i at time t, defined as $C_i^{(t)} = \frac{1}{\sigma_i^{2,(t)}} \cdot \exp\left(-\frac{\mathrm{Var}(\mathbf{w}^\top \tilde{\mathbf{f}}_i^{(t)})}{\xi}\right)$.

Exponential Smoothing A time series smoothing technique applied to feature vectors: $\tilde{\mathbf{f}}_i^{(t)} = \alpha \cdot \mathbf{f}_i^{(m_t)} + (1 - \alpha) \cdot \tilde{\mathbf{f}}_i^{(t-1)}$.

Blending Parameter λ The weight controlling the balance between Bayesian skill estimates and direct feature contributions in the final rating: $R_i^{(t)} = \lambda \cdot \mu_i^{(t)} + (1 - \lambda) \cdot \mathbf{w}^\top \tilde{\mathbf{f}}_i^{(t)}$.

AUC-ROC (Area Under the ROC Curve) A measure of binary classification performance across all decision thresholds, ranging from 0.5 (random) to 1.0 (perfect).

Spearman's Rank Correlation ρ A nonparametric measure of rank correlation between two variables, used to compare system rankings with expert judgments.

Gini Coefficient A measure of statistical dispersion, used here to quantify the separation between player skill ratings (higher values indicate better discrimination).

Rating Volatility The average absolute change in a player's rating per match, calculated as $\frac{1}{M_i} \sum_{t=1}^{M_i} |R_i^{(t)} - R_i^{(t-1)}|$.

Rank Flip Rate The percentage of player pairs whose relative ranking changes between consecutive rating updates, excluding pairs with small rating differences (< 10 points).

Demofile The complete recording of a CS2 match containing timestamped logs of all in-game events, from which our raw feature vectors are extracted.

Smurfing The practice of experienced players competing on new or secondary accounts, creating a mismatch between apparent and actual skill level.

KAST An existing CS2 statistic: percentage of rounds where player gets a Kill, Assist, Survives, or gets Traded. Part of HLTV Rating 2.0 but refined in our system with contextual weighting.

ADR (Average Damage per Round) Total damage dealt divided by rounds played. In our system, damage is further decomposed by context (armor vs. health, eco vs. gun rounds).

Impact Rating An existing HLTV metric measuring multi-kill rounds and opening kills. Our system generalizes this concept through feature decomposition.

Appendix B Hyperparameter Configurations

This section provides the complete hyperparameter configurations for CS2 Rating 3.0 and baseline models. All values were determined through grid search with 5-fold cross-validation on the training set $\mathcal{D}_{\text{train}}$.

B.1 CS2 Rating 3.0 Hyperparameters

Table 7: CS2 Rating 3.0 hyperparameter values with search ranges and optimal selections.

Parameter	Symbol	Search Range	Optimal Value
Feature Contribution Decomposition Module (LightGBM)			
Number of trees	T	$\{100, 300, 500, 700, 1000\}$	500
Maximum tree depth	D	$\{4, 6, 8, 10, 12\}$	8
Learning rate	η_{GBDT}	$\{0.01, 0.05, 0.1, 0.2\}$	0.05
Minimum child samples	$m_{\min}$	$\{20, 50, 100, 200\}$	50
Feature fraction	f	$\{0.6, 0.7, 0.8, 0.9\}$	0.8
Bagging fraction	b	$\{0.6, 0.7, 0.8, 0.9\}$	0.8
Early stopping rounds	e	$\{50, 100, 150\}$	100
Bayesian Pairwise Comparison Engine			
Scale parameter	η	$\{0.8, 1.0, 1.2, 1.4, 1.6\}$	1.2
Volatility parameter	τ	$\{0.1, 0.15, 0.2, 0.25, 0.3\}$	0.15
Prior mean	μ_0	$\{1000, 1200, 1500\}$	1200
Prior variance	σ_0^2	$\{200, 400, 600\}$	400
VI iterations	I_{VI}	$\{1000, 2000, 3000, 4000\}$	2000
Learning rate (VI)	η_{VI}	$\{0.001, 0.003, 0.01, 0.03\}$	0.001
Temporal and Integration Parameters			
Smoothing factor	α	$\{0.1, 0.2, 0.3, 0.4, 0.5\}$	0.3
Blending parameter	λ	$\{0.5, 0.6, 0.7, 0.8, 0.9\}$	0.7
Confidence normalization	ξ	$\{1, 2, 5, 10\}$	2
Moving window size	W	$\{10, 20, 30, 50\}$ matches	30

B.2 Baseline Model Hyperparameters

Table 8 shows the hyperparameter configurations for baseline models, optimized using the same grid search procedure.

Table 8: Hyperparameter configurations for baseline rating systems.

Model	Parameter	Range	Optimal
HLTV Rating 2.0	(Fixed linear weights)	–	(Standard formula)
Elo	K-factor	$\{16, 24, 32, 40, 48\}$	32
Elo	Logistic scale	$\{200, 400, 600, 800\}$	400
Glicko-2	τ (volatility)	$\{0.3, 0.5, 0.7, 0.9\}$	0.5
Glicko-2	ϵ (convergence)	$\{10^{-6}, 10^{-5}, 10^{-4}\}$	10^{-6}
TrueSkill2	β (performance variance)	$\{4.17, 8.33, 16.67\}$	8.33
TrueSkill2	τ (dynamics factor)	$\{0.1, 0.3, 0.5, 0.7\}$	0.3
GBDT-Direct	Number of trees	$\{200, 400, 600, 800\}$	400
GBDT-Direct	Max depth	$\{6, 8, 10, 12\}$	10
GBDT-Direct	Learning rate	$\{0.01, 0.05, 0.1\}$	0.05

B.3 Hyperparameter Sensitivity Analysis

Figure 10 shows the sensitivity of CS2 Rating 3.0's performance to variations in key hyperparameters. The system demonstrates robustness within reasonable ranges of each parameter, with performance degrading gracefully outside optimal regions.

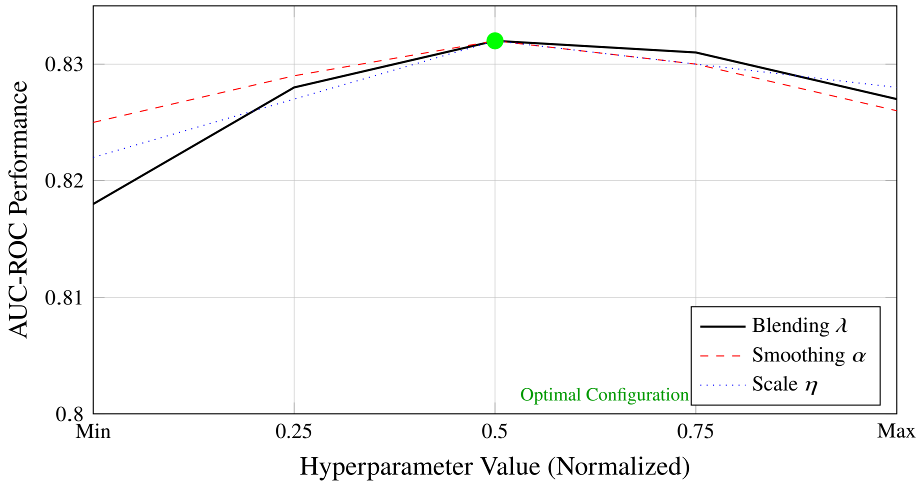

Figure 10: Sensitivity of CS2 Rating 3.0 performance to variations in key hyperparameters. Values are normalized to [0,1] across search ranges. The system shows robustness with performance within 1% of optimal across most of the parameter space.

The hyperparameter optimization process required approximately 120 GPU-hours on NVIDIA RTX A6000 hardware. The final configuration achieves a balance between model complexity and generalization performance, as evidenced by consistent performance on the validation and test sets.

(this poster is chosen by authors.)